An Educational Tool For Numerical Analysis

WELCOME | ERROR | BISECTION METHOD | NEWTON RAPHSON METHOD | SECANT METHOD | SPLINE INTERPOLATION | TRAPEZOIDAL RUL

Function given x^2-2*x+1 e.g.: x^2-2*x+1

Insert the interval [a, b]

a 1 **b** 2

Bisection Tolerance 0.0001 e.g.: 0.0001

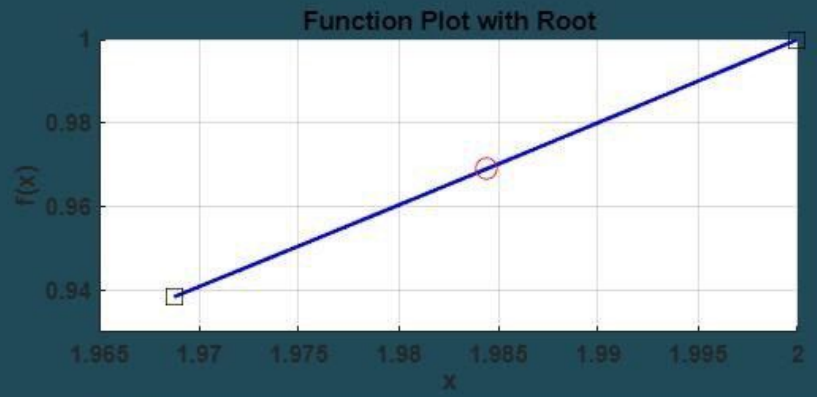
level of error that is acceptable

Bisection Max iteration 5 maximum iteration can be done

Compute Root

ANSWER:

root bisection 1.984 **Bisection Iteration** 5



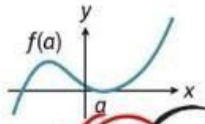
Bisection Data

Iter	a	b	c	f(c)
1	1.000000	2.000000	1.500000	0.250000
2	1.500000	2.000000	1.750000	0.562500
3	1.750000	2.000000	1.875000	0.765625
4	1.875000	2.000000	1.937500	0.878906
5	1.937500	2.000000	1.968750	0.938477

WELCOME TO PART 2:
BISECTION METHOD

Find the root of an equation $f(x) = 0$

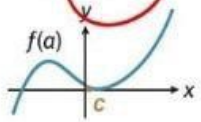
1. Choose interval $[a, b]$



$$c = \frac{a+b}{2}$$

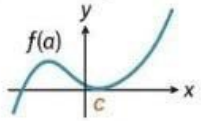
2. Find midpoint

middle point between point a and point b



If $f(c) = 0$
- Root is c

3. Check sign of f(a) and f(b)



The method always works

The Bisection Method is a way to find the root(solution) of an equation

GUIDE:

STEP 1: Enter the function that you want
STEP 2: Choose two points [a,b]
STEP 3: Set the tolerance (the accuracy you want for the answer)
STEP 4: Set the maximum number of iterations to control how many steps the method will take
STEP 5: Finally, press the Compute Button and the the root will be shown

"Cut, Divide, Conquer. The Bisection Method finds root by halving the problem step by step!"

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WELCOME | ERROR | BISECTION METHOD | NEWTON RAPHSON METHOD | SECANT METHOD | SPLINE INTERPOLATION | TRAPEZOIDAL RUL

Insert the function

Function f(x) x.^3+3*x+1 Exp: x.^3 +3*x +1

Initial guess (x_n) 0.5

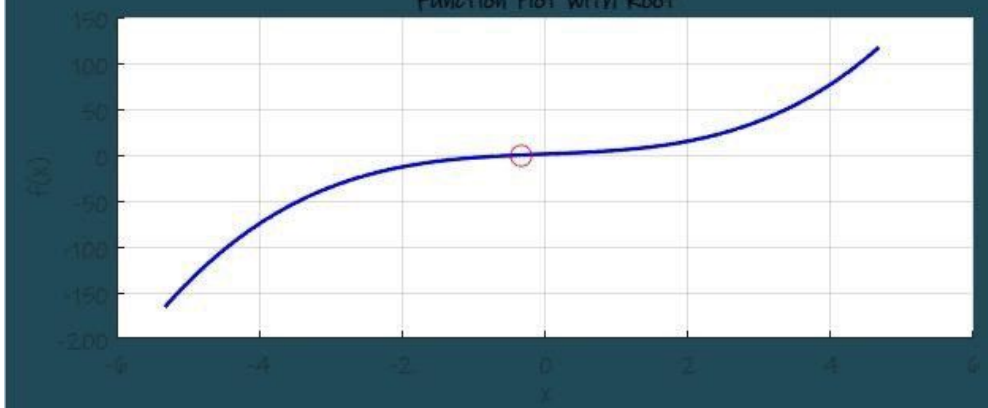
Error tolerance 0.0004 Level of error that is acceptable

Max Iterations 6 Maximum iteration can be done

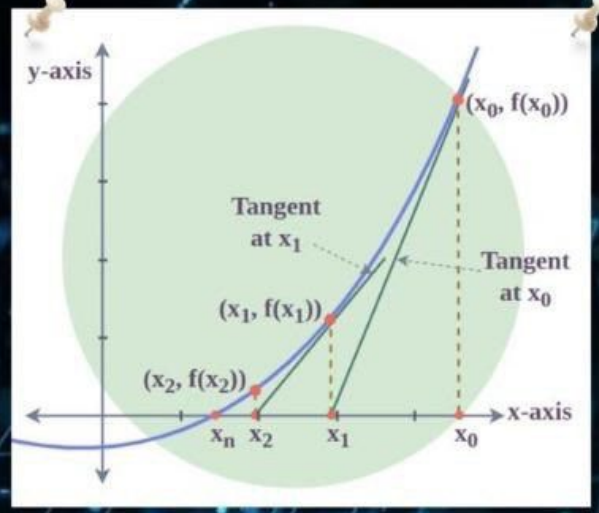
Compute **Answer** x = -0.322185 (found at iteration 4)

Iteration	x_n	f(x_n)	f'(x_n)	x_{n+1}	x_{n+1} - x_n
1	0.5000	2.6250	3.7500	-0.2000	0.7000
2	-0.2000	0.3920	3.1200	-0.3256	0.1256
3	-0.3256	-0.0115	3.3181	-0.3222	0.0035
4	-0.3222	-0.0000	3.3114	-0.3222	0.0000

Function Plot with Root



WELCOME TO PART 3:
NEWTON RAPHSON METHOD



Newton Raphson method is to find the root (solution) of an equation: f(x) = 0 That means finding the value of x where the function crosses the x-axis.

FORMULA: $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

Where:

- x_n = current approximation
- $f'(x_n)$ = derivative of $f(x)$
- x_{n+1} = next (improved) approximation

ADVANTAGES !!!

- Converges very fast
- Simple formula and easy to apply
- Suitable for nonlinear equations

GUIDE:

STEP 1: Enter the function that you want
STEP 2: Choose initial guess
STEP 3: Set the tolerance (the accuracy you want for the answer)
STEP 4: Set the maximum number of iterations to control how many steps the method will take
STEP 5: Finally, press the Compute Button and the root will be shown

"Guess, Differentiate, Repeat Until Root Appears!"

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WELCOME | ERROR | BISECTION METHOD | NEWTON RAPHSON METHOD | SECANT METHOD | SPLINE INTERPOLATION | TRAPEZOIDAL RUL

Function x^2-2*x+1 e.g.: x^2-2x+1

Insert the initial guess :

x0 0.5 **x1** 2.9

x0 and x1 can't be 0

level of error that is acceptable

Tolerance 0.0001 e.g.: 0.0001

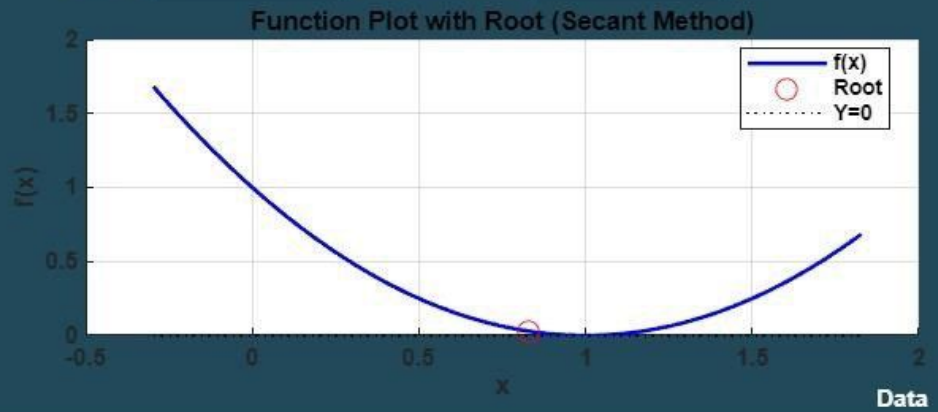
maximum iteration

Max iteration 5

Compute

ANSWER:

Root 0.8273 **Iteration** 5



Data

Iter	x(n-1)	x(n)	f(x(n-1))	f(x(n))	x(n+1)	|x(n+1)-x(n)|
1	0.5000000	2.9000000	0.2500000	3.6100000	0.3214286	2.5785714
2	2.9000000	0.3214286	3.6100000	0.4604592	-0.0555556	0.3769841

WELCOME TO PART 4: SECANT METHOD

Secant Method – Derivation

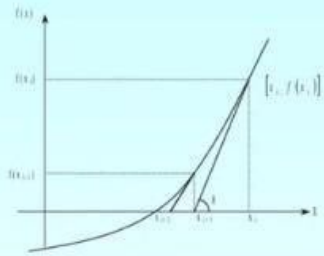


Figure 1 Geometrical illustration of the Newton-Raphson method.

Newton's Method

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)} \quad (1)$$

Approximate the derivative

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} \quad (2)$$

Substituting Equation (2) into Equation (1) gives the Secant method

$$x_{i+1} = x_i - \frac{(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})} f(x_i)$$

The Secant Method is a numerical to find the root of a function f(x) = 0, it uses two initial guesses

GUIDE:

STEP 1: Enter the function f(x)
STEP 2: Insert two intial guesses x0 and x1
STEP 3: Set the tolerance and maximum iterations
STEP 4: Press Compute to run the Secant Method
STEP 5: View the root, iteration details and graph of the function with secant lines

"Guess, Connect, Converge - the Secant Way."

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WELCOME | ERROR | BISECTION METHOD | NEWTON RAPHSON METHOD | SECANT METHOD | SPLINE INTERPOLATION | TRAPEZOIDAL RUL

Insert the X and Y values

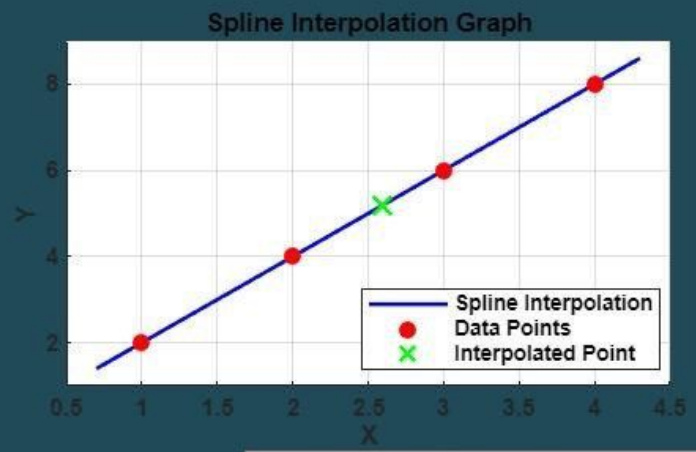
X Values 1,2,3,4 e.g.: 1, 2, 3 or 1 2 3

Y Values 2,4,6,8 e.g.: 1, 2, 3 or 1 2 3

Find y at x

X Interp 2.59 x must be in between X Values range

Interpolate



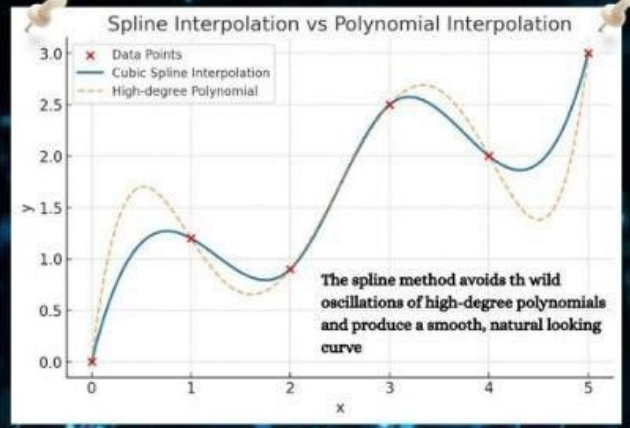
Spline Interpolation Data

--- Spline Interpolation Results ---
Interpolated Y for X = 2.59 is: 5.18

Input Data:
X = [1 2 3 4]
Y = [2 4 6 8]

Note: Spline interpolation produces a piecewise polynomial. No single equation is generated.

WELCOME TO PART 5: SPLINE INTERPOLATION



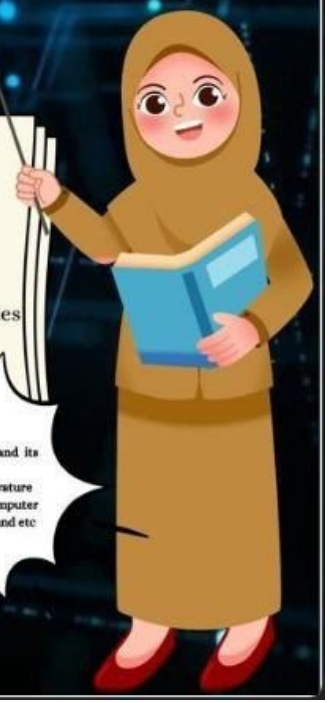
Spline Interpolation is a method of constructing a smooth curve that passes through a given set of point (called knots)

GUIDE:

STEP 1: Enter data points (X and Y)
STEP 2: Enter Target X
(type the x-value where you want to find the missing y)
⚠ Make sure this x is between the range of your X values
STEP 3: Press "Interpolate"
STEP 4: View result & graph

ADVANTAGES!!!
- Can produces smooth curve
- Can model complex shapes and its accurate
- Have continous slope and curvsture
- Used in industries: Computer graphics, engineering design and etc

"Your data, prefectly curved."



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ERROR | BISECTION METHOD | NEWTON RAPHSON METHOD | SECANT METHOD | SPLINE INTERPOLATION | TRAPEZOIDAL RULE | MIDPOIN

Insert the function

f(x) x.^2+3*x-1 exp: x.^2 + 3*x -3

The start of the interval

Lower limit, a 1

The end of the interval

Upper limit, b 2

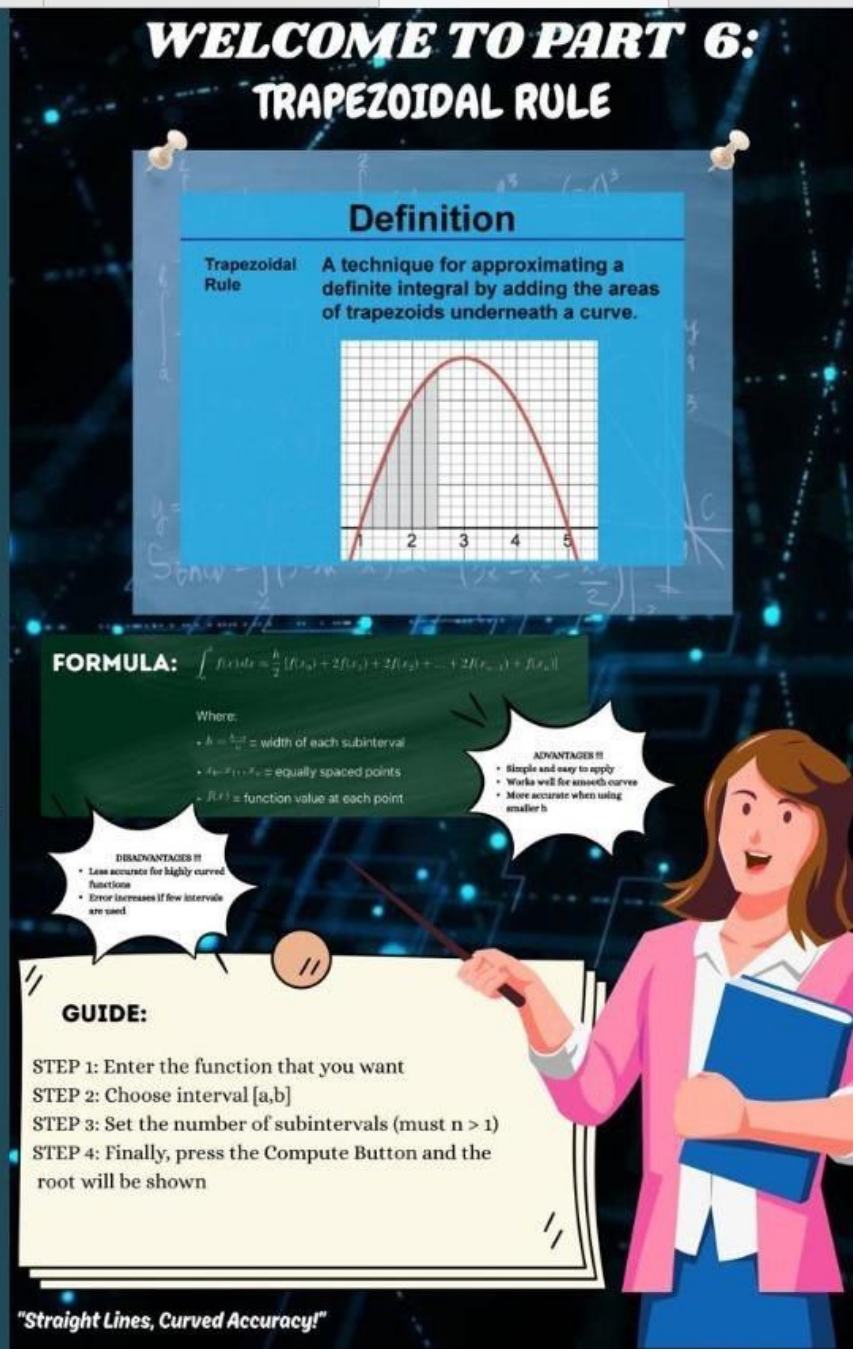
Number of subintervals:
Basic: n = 1
Composite: n > 1 **n** 3

Compute

Results

Method: Trapezoidal Rule
Numerical Integral ≈ 5.851852
Exact Integral = 5.833333
Absolute Error = 0.018519

Numerical Integration (Trapezoidal Rule)



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BISECTION METHOD | NEWTON RAPHSON METHOD | SECANT METHOD | SPLINE INTERPOLATION | TRAPEZOIDAL RULE | MIDPOINT RULE

Insert the function

f(x) 4-x.^2+3*x-3 exp: 4-x.^2 + 3*x -3

The start of the interval

Lower limit, a 1

The end of the interval

Upper limit, b 2

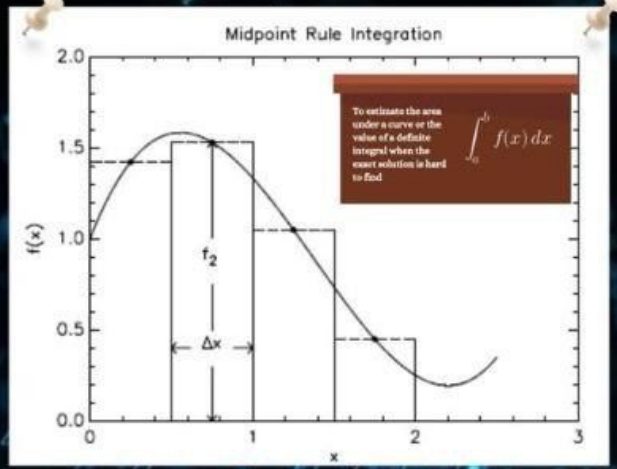
Number of subintervals:
Basic: n = 1
Composite: n > 1 **n** 2

Compute

Results

Method: Composite Midpoint Rule
Numerical Integral ≈ 3.187500
Exact Integral = 3.166667
Absolute Error = 0.020833

WELCOME TO PART 7:
MIDPOINT RULE



FORMULA: $\int_a^b f(x)\,dx \approx h\sum_{i=1}^{n} f\left(\frac{x_{i-1}+x_i}{2}\right)$

Where:

- $h = \frac{b-a}{n}$ → width of each subinterval
- $\frac{x_{i-1}+x_i}{2}$ → midpoint of each subinterval

ADVANTAGES !!
- Simple and efficient
- Usually more accurate than the trapezoidal rule for smooth curve
- Reduces some types of error because midpoint balances the curve

GUIDE:

STEP 1: Enter the function that you want
STEP 2: Choose interval [a,b]
STEP 3: Set the number of subintervals (must n > 1)
STEP 4: Finally, press the Compute Button and the root will be shown

"Find the Middle, Get the Best Estimate!"

An Educational Tool For Numerical Analysis

NEWTON RAPHSON METHOD | SECANT METHOD | SPLINE INTERPOLATION | TRAPEZOIDAL RULE | MIDPOINT RULE | SIMPSON RULE | EULE

Insert the function

f(x) x.^2+3*x-3 exp: x.^2 + 3*x -3

The start of the interval

Lower limit, a 1

The end of the interval

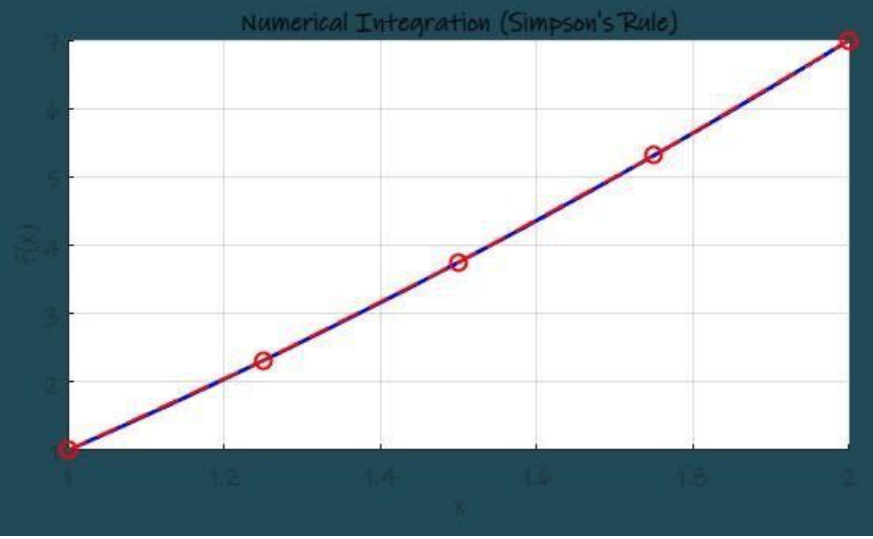
Upper limit, b 2

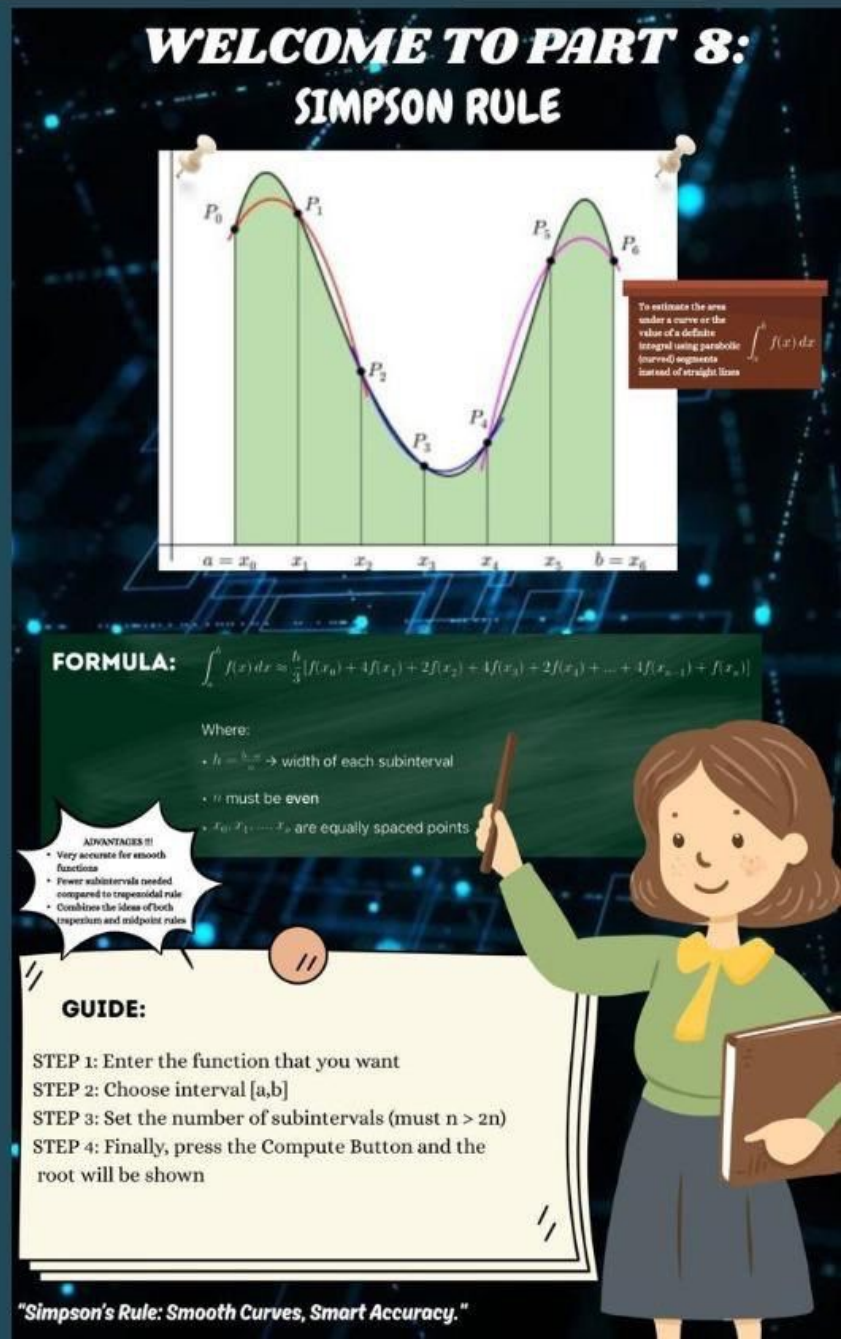
Number of subintervals:
Basic: n = 2
Composite: n > 2n **n** 4

Compute

Results

Method: Simpson's Rule
Numerical Integral ≈ 3.833333
Exact Integral = 3.833333
Absolute Error = 0.000000





An Educational Tool For Numerical Analysis

< NTON RAPHSON METHOD | SECANT METHOD | SPLINE INTERPOLATION | TRAPEZOIDAL RULE | MIDPOINT RULE | SIMPSON RULE | EULER

Example: t+y-1

Notes:
Use differential eqn

Function t+y-1

Initial time t0 0.5

Initial value y0 0.8

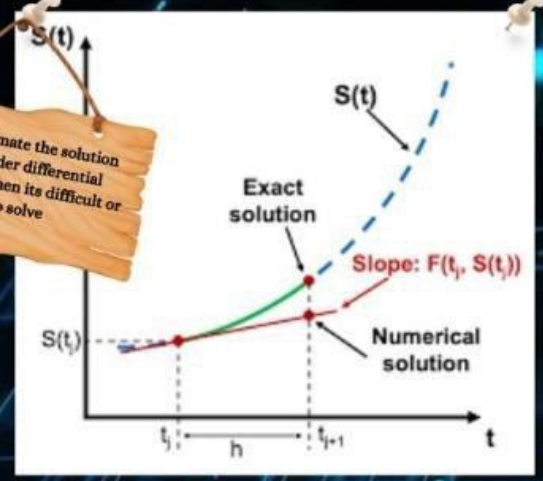
Step size h 3

Number of steps n 4

Press

Approximate 320.3

WELCOME TO PART 9: EULER METHOD



FORMULA: $y_{i+1} = y_i + h\, f(x_i, y_i)$

ADVANTAGES !!
- Easy and simple to apply
- Good for quick numerical estimates

Where:

- $x_{i+1} = x_i + h$
- h = step size (interval width)
- $f(x_i, y_i)$ = slope of the curve at (x_i, y_i)

GUIDE:

STEP 1: Enter the differential function that you want in term of t and y
STEP 2: Choose intial time, t0 and initial value, yo
Step 3: Set the step size, h
STEP 3: Set the number of steps, n
STEP 4: Finally, press the "Press" Button and the root will be shown

"Euler Method: Small Steps, Big Solutions."